



Raritone Summer Internship 2026

Research on AI-Powered Fashion Technologies

Introduction

The fashion industry is increasingly adopting Artificial Intelligence to enhance customer experiences and improve digital shopping platforms.

This internship focused on researching various AI-driven fashion technologies including:

- Computer Vision
- Human Pose Estimation
- Segmentation & Human Parsing
- 2D Virtual Try-On Systems
- 3D Virtual Try-On Systems
- Fashion Recommendation Engines
- AR/VR Fashion Technologies

Objectives

- Study AI applications in fashion technology.
- Explore Virtual Try-On systems.
- Understand body analysis techniques.
- Research recommendation systems.
- Analyze fashion datasets.
- Investigate AR/VR integration.
- Document findings and observations.



OBJECTIVES

- 1. Study AI Applications in Fashion Technology**
Explore the role of AI, Computer Vision and Machine Learning in modern fashion platforms.
- 2. Explore Virtual Try-On Systems**
Research 2D and 3D virtual try-on techniques for realistic garment visualization.
- 3. Understand Body Analysis Techniques**
Investigate body detection, pose estimation, landmark identification and body measurements.
- 4. Research Recommendation Systems**
Study content-based, collaborative and hybrid recommendation approaches.
- 5. Analyze Fashion Datasets**
Explore datasets used for training and evaluating fashion AI models.
- 6. Investigate AR/VR Integration**
Examine the use of AR and VR to create immersive fashion shopping experiences.
- 7. Document Findings and Observations**
Prepare detailed reports, presentations and documentation of research outcomes.

Fashion AI Ecosystem

User Image
↓
Body Detection
↓
Pose Estimation
↓
Segmentation
↓
Virtual Try-On
↓
Recommendation Engine
↓
Personalized Output



Body Detection & Pose Estimation

Technologies Studied

- MediaPipe
- OpenPose
- OpenCV

Key Functions

- Human Detection
- Landmark Identification
- Skeleton Tracking
- Body Analysis

Findings

- Accurate landmarks improve fitting quality.
- Pose estimation supports garment alignment.



Segmentation & Human Parsing

Technologies Studied

- U2Net
- Detectron2

Applications

- Cloth Segmentation
- Human Parsing
- Background Removal

Findings

- Better segmentation improves virtual try-on accuracy.

2D Virtual Try-On Systems

Workflow

User Image



Pose Detection



Segmentation



Garment Alignment



Virtual Try-On Output

Models Explored

VITON-HD

HR-VITON

3D Virtual Try-On Systems

Concepts Studied

- Avatar Generation
- Body Reconstruction
- Cloth Simulation

Technologies

- Blender
- Unity
- Three.js

Findings

- More immersive than traditional systems
- Better personalization





Recommendation Engine Research

Approaches Studied

Content-Based Filtering
Collaborative Filtering
Hybrid Recommendation Systems

Benefits

Personalized suggestions
Better product discovery
Enhanced user engagement

Datasets Explored

Datasets Studied

- DeepFashion
- DeepFashion2
- VITON-HD

Importance

- Model Training
- Evaluation
- Performance Improvement

Observation

Higher-quality datasets produce more accurate AI systems.

Challenges & Solutions

Challenge

Complex Fashion
Technologies

Understanding AI Models

Dataset Analysis

Workflow Integration

Solution

Conducted detailed
research

Reviewed documentation
and papers

Compared multiple
datasets

Created structured
architecture diagrams

Future Enhancements

- Real-Time Virtual Try-On AI
- Fashion Assistants
- Advanced Recommendation Engines
- AR Shopping Experiences
- Enhanced 3D Avatars
- Realistic Cloth Simulation



Key Learnings

- Understanding of Computer Vision concepts.
- Knowledge of Virtual Try-On architectures.
- Familiarity with Recommendation Systems.
- Exposure to AR/VR fashion technologies.
- Improved research and documentation skills.
- Better understanding of AI workflows.



Conclusion

The internship provided valuable insights into Artificial Intelligence applications in the fashion industry. Through extensive research on Virtual Try-On Systems, Recommendation Engines, Computer Vision, and AR/VR technologies, a deeper understanding was developed regarding how AI can transform digital fashion experiences.





Thank you!

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